Supervised Learning Assignment Proposal

Project Title

Customer Purchase Behavior Prediction using Machine Learning

Dataset Details

- **Dataset Name:** Online Retail II UCI Dataset
- **Source:** Kaggle
- **Dataset Link:** https://www.kaggle.com/datasets/mashlyn/online-retail-ii-uci
- **Number of Rows:** ~1,000,000+ transactions
- **Number of Columns:** 8

Key Columns:

- Invoice
- StockCode
- Description
- Quantity
- InvoiceDate
- Price
- Customer ID
- Country

Target Variable

HighValue Transaction (Binary Classification)

- 1 → High-value transaction (above median total price)
- 0 → Low-value transaction

Type of Task

Classification Problem

Problem Statement

Businesses aim to understand customer purchasing behavior to improve revenue and marketing strategies. This project focuses on analyzing retail transaction data to identify key factors influencing high-value purchases and building a machine learning model to predict such transactions.

Research Questions

1. Does the quantity of items purchased affect total revenue?
2. Which countries generate the highest revenue?
3. How does product price influence purchasing behavior?
4. Which products contribute the most to total revenue?
5. Are there seasonal trends in sales over time?
6. What features most influence transaction value?
7. Can machine learning accurately predict high-value transactions?

Proposed Methodology

Step 1: Data Collection

- Import dataset from Kaggle into Kaggle Notebook

Step 2: Data Preprocessing

- Handle missing values using `dropna()`

- Clean column names
- Convert date column to datetime format

Step 3: Feature Engineering

- Create new feature:
    - **TotalPrice = Quantity × Price**
- Create target variable:
    - HighValue (binary classification)

Step 4: Exploratory Data Analysis (EDA)

- Scatter plots
- Bar charts
- Time-series analysis

Step 5: Model Building

- Split dataset into training and testing sets
- Train machine learning models

Step 6: Evaluation

- Evaluate model performance using classification metrics

Step 7: Output Generation

- Save figures as PDF
- Save tables as CSV

Machine Learning Models Used

- Random Forest Classifier
- Random Forest Regressor (for feature importance analysis)

Evaluation Metrics

- Accuracy
- Confusion Matrix
- Precision
- Recall
- F1-score

Expected Figures

1. Scatter plot (Quantity vs Revenue)
2. Bar chart (Revenue by Country)
3. Scatter plot (Price vs Quantity)
4. Bar chart (Top Products by Revenue)
5. Line chart (Monthly Sales Trend)
6. Feature Importance plot
7. Confusion Matrix

Expected Tables

1. Quantity vs Revenue data
2. Revenue by Country
3. Price vs Quantity
4. Top Products
5. Monthly Sales
6. Feature Importance values
7. Actual vs Predicted values

Conclusion

This project will provide insights into customer purchasing behavior and demonstrate the effectiveness of machine learning techniques in predicting high-value transactions. The results can help businesses make data-driven decisions to improve sales and marketing strategies.

Tools & Technologies

- Python
- Pandas
- Matplotlib
- Seaborn
- Scikit-learn
- Kaggle Notebooks